

JELIN RAPHAEL AKKARA

Summary: Pursuing PhD in Neuro-Symbolic Approaches to Embodied AI. Has experience in Computer Vision, Embodied AI, and Vision-Language Models. Interested in reasoning models, representation learning.

Email: jelinraphaelakkara@gmail.com

Portfolio: <https://jelinr.github.io/>

LinkedIn: [linkedin.com/in/jelin-raphael-akkara](https://www.linkedin.com/in/jelin-raphael-akkara)

GitHub: <https://github.com/JelinR>

CERTIFICATIONS

Language Proficiency in English (2022)

IELTS Band 8, CEFR Level C1

Google Data Analytics Professional Certificate (2022)

Statistical Data Analysis with SQL, R, Tableau

SKILLS

Programming Languages

Python, R, SQL

Programming Frameworks

Habitat AI, Hydra, Dask, PySpark, Docker, LangChain, HuggingFace

Programming Libraries

TensorFlow, Keras, PyTorch, Scikit-Learn, NLTK, spaCy, OpenCV, YOLO, Detectron2

Languages

English (native), Malayalam (native), Hindi (intermediate), Italian (beginner)

INITIATIVES

Venice International Workshop, 'Physics of Data', Co-Organizer

Held on 23-24 May, 2024

TEDxCalicut Co-Organizer (2019)

Managed guest logistics and communication.

RESEARCH EXPERIENCE

GenAI Research Intern

ViWear

Enhanced Stable Diffusion-derived virtual try-on model by 7-12% (perceptual metrics), and lowered the inference time (1 fps to 3 fps). This was done using architectural improvements and fine-tuning.

Computer Vision Research Intern

Fondazione Bruno Kessler (FBK) / VIMP, University of Padova

Designed a modular framework for efficient Visual SLAM mapping (2D Semantic Map, VLFM). Developed a grid-based multi-modal embedding mapping, reducing memory usage by 99.75% relative to dense mappings.

EDUCATION

Masters in Physics of Data

University of Padua, Italy

GPA : 110/110

2022 - 2025

Thesis: Zero-Shot Object Goal Navigation using Online Image Retrieval

Relevant Coursework: Vision and Cognitive Systems (Computer Vision), Reinforcement Learning, Human Data Analytics, Natural Language Processing, Machine Learning

B.Tech in Engineering Physics

National Institute of Technology, Calicut, India

GPA : 8.43 / 10

2018 - 2022

First Class with Distinction

PROJECTS

Lightweight CNN for Speech Keyword Spotting

Designed a novel lightweight CNN (32k parameters) achieving 90% accuracy, and 37ms inference, rivaling SoTA models like TDNN (250k parameters, 94% accuracy).

YOLOv8n Object Detection using Blob Enhancers

Enhanced YOLOv8n for small human detection (far away or occluded persons) by 1.1%, with a minor preprocessing speed increase of 2 ms (7 ms to 9 ms).

Learning Immanuel Kant using LLM and RAG

Tested RAG with Llama-2, FAISS, and HuggingFace for semantic understanding of philosophical texts by optimizing prompt engineering.